

Banking & Financial Glossary

ABIX Data Explorer — Financial Metrics, Ratios, Algorithms & Banking Concepts in Algeria

This document provides the comprehensive methodological reference for all accounting aggregates, formulas, and statistical methods implemented in the ABIX Data Explorer platform across the 21 commercial banks in Algeria.

Scope: 21 Commercial Banks in Algeria	Indexed Terms: 52 concepts
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FINANCIAL & BALANCE SHEET INDICATORS

Total Assets (Balance Sheet Total) (TB)

Unit: Mds DZD

Measures the overall financial size of a bank by aggregating all its assets and uses of funds.

Total Assets represents the sum of all asset items on the balance sheet as of December 31 (cash, securities, customer loans, fixed assets). In ABIX, it serves as the primary metric for measuring bank size and market share.

ABIX FORMULA:
Total Actifs = Caisse & Banques centrales + Titres & Bons du Trésor + Crédits clientèle + Autres actifs

Interpretation: A high balance sheet indicates market scale and intermediation capacity within the banking system.

Concrete example: The aggregate balance sheet of the 21 banks reaches around 22,000 billion DZD, with nearly 88% held by public banks.

ABIX Modules: Vue Secteur, Classements, Profil Banque, Historique, Parts de marché, Executive Dashboard

Customer Deposits (DEP)

Unit: Mds DZD | ▲ Higher = Better

Total funds entrusted to the bank by households, corporates, and institutions as demand, term, or savings deposits.

As the primary liability resource, customer deposits measure deposit collection capacity and customer trust. In ABIX, multi-year deposit growth is compared against loans to assess liquidity dynamics.

ABIX FORMULA:
Dépôts Totaux = Dépôts à vue + Dépôts à terme + Comptes d'épargne + Autres dépôts clientèle

Interpretation: Steady deposit growth strengthens bank funding stability without excessive reliance on money markets.

Concrete example: In 2022-2023, customer deposits grew by +6.2% exceeding 14,600 billion DZD.

ABIX Modules: Vue Secteur, Classements, Profil Banque, Historique, Parts de marché, Comparateur

Customer Loans (Gross Loans) (CRED)

Unit: Mds DZD

Total gross amount of financings, overdrafts, and credit facilities extended to households, businesses, and public entities.

Customer loans represent the core earning asset of a universal bank. ABIX monitors lending growth, transformation ratios, and relative financing to the real economy.

ABIX FORMULA:
Crédits Bruts = Crédits d'exploitation + Crédits d'investissement + Crédits immobiliers + Crédits aux particuliers

Interpretation: Reflects direct financing of economic activity and generates the majority of interest revenues.

Concrete example: Total loan stock to the economy reached approx. 8,840 billion DZD in 2023, with an overall transformation ratio of ~60%.

ABIX Modules: Vue Secteur, Classements, Profil Banque, Historique, Parts de marché, Comparateur

Net Banking Income (NBI) (PNB / NBI)

Unit: Mds DZD | ▲ Higher = Better

Equivalent to a bank's net operating revenue, calculated as net interest margin plus net fee income and trading gains.

Net Banking Income measures the gross revenue generated across all banking and intermediation operations. It serves as the denominator for the Cost-to-Income ratio and operational efficiency metrics.

ABIX FORMULA:

PNB = (Produits d'intérêts - Charges d'intérêts) + (Commissions perçues - Commissions payées) + Gains/pertes de change et titres

Interpretation: Consistent NBI growth highlights commercial vitality and balanced asset/service pricing.

Concrete example: BEA generated 112.4 billion DZD of NBI, ranking 2nd nationally behind BNA.

ABIX Modules: [Vue Secteur](#), [Classements](#), [Profil Banque](#), [Historique](#), [Parts de marché](#), [Comparateur](#), [Scores financiers](#)

Gross Operating Income (GOI) (RBE / GOI)

Unit: Mds DZD | ▲ Higher = Better

Difference between NBI and operating expenses (personnel costs, admin overhead, depreciation).

GOI measures pure operational profitability prior to any loan impairment charges (cost of risk) or tax expenses.

ABIX FORMULA:

RBE = Produit Net Bancaire - Frais de personnel - Autres charges générales d'exploitation - Dotations aux amortissements

Interpretation: A growing positive GOI shows revenues comfortably absorb infrastructure, branch, and staff operating expenses.

Concrete example: For an NBI of 100M DZD and 40M DZD operating expenses, GOI stands at 60M DZD.

ABIX Modules: [Vue Secteur](#), [Classements](#), [Profil Banque](#), [Historique](#)

Net Profit (Net Income) (RN / NI)

Unit: Mds DZD | ▲ Higher = Better

Final profit or loss remaining after deducting all expenses, loan loss provisions, and corporate income taxes.

The bottom-line figure of the income statement, Net Profit serves to pay dividends to shareholders and strengthen regulatory equity capital via retained earnings.

ABIX FORMULA:

Résultat Net = RBE - Dotations nettes aux provisions pour créances + Produits/charges exceptionnels - Impôts sur les bénéfices (IBS)

Interpretation: Measures overall profit generation and forms the numerator for Return on Equity (ROE) and Return on Assets (ROA).

Concrete example: Aggregate banking sector net income increased by +14.8% reaching 225.6 billion DZD in 2023.

ABIX Modules: [Vue Secteur](#), [Classements](#), [Profil Banque](#), [Historique](#), [Scores financiers](#), [Executive Dashboard](#)

Total Equity (Shareholders' Funds) (FP / EQ)

Unit: Mds DZD | ▲ Higher = Better

Permanent capital belonging to shareholders, including share capital, retained earnings, reserves, and current profit.

Shareholders' Equity acts as the primary solvency buffer absorbing potential loan losses. In ABIX, it is evaluated through the Equity-to-Assets ratio and leverage capacity.

ABIX FORMULA:

Fonds Propres = Capital Social + Primes d'émission + Réserves légales et statutaires + Report à nouveau + Résultat de l'exercice

***Interpretation:** High equity cushions ensure resilience against macroeconomic shocks and enable financing larger syndications.*

Concrete example: The median Equity-to-Assets ratio across Algerian banks stands around 8.3%.

ABIX Modules: Vue Secteur, Classements, Profil Banque, Historique, Scores financiers, Quatre Dimensions

Treasury Bills & Sovereign Securities (BDT / OAT)

Unit: Mds DZD

Sovereign debt securities issued by the Algerian State and held as assets by commercial banks.

Treasury bills constitute a major asset holding for Algerian banks, especially state-owned banks. They provide secure interest yields with zero regulatory risk-weighting.

ABIX FORMULA:

BDT = Bons du Trésor à court terme (BTCT) + Bons du Trésor assimilables (OAT)

***Interpretation:** High sovereign bond holdings boost liquidity and regulatory ratios but may yield lower returns than private corporate lending.*

Concrete example: State-owned banks hold the vast majority of the sovereign bond portfolio in the domestic market.

ABIX Modules: Vue Secteur, Profil Banque, Comparateur

Off-Balance Sheet Commitments (HB / OBS)

Unit: Mds DZD

Financial commitments given or received by the bank not immediately recorded on the balance sheet (guarantees, letters of credit).

Off-balance sheet operations capture foreign trade financing and public procurement support (bid bonds, performance bonds), generating fee income without upfront cash outlay.

ABIX FORMULA:

Engagements Hors-Bilan = Cautions et avals donnés + Engagements de financement accordés + Crédits documentaires ouverts

***Interpretation:** High off-balance sheet volume signals strong positioning in trade finance and major infrastructure projects.*

Concrete example: BEA and BNA record massive off-balance sheet exposures due to their central role in oil and industrial import-export transactions.

ABIX Modules: Profil Banque, Comparateur, Classements

RATIOS & PROFITABILITY

Return on Equity (ROE) (ROE / RCP)

Unit: % | ▲ Higher = Better

Measures a bank's ability to generate net income per unit of equity capital invested by shareholders.

ROE is the benchmark metric for financial profitability. In ABIX, it is computed on closing equity as well as average equity (ROAE) to smooth capital variations during the financial year.

ABIX FORMULA:

$$\text{ROE} = (\text{Résultat Net} / \text{Fonds Propres Comptables}) \times 100$$

Interpretation: An ROE exceeding 12-14% is considered strong performance in the Algerian banking environment.

Concrete example: With an ROE of 14.8%, BEA and Al Baraka (16.4%) rank in the top quartile (Q4) for equity returns.

ABIX Modules: Vue Secteur, Classements, Profil Banque, Matrice Stratégique, Heatmap, Scores financiers, Quatre Dimensions

Return on Average Equity (ROAE) (ROAE)

Unit: % | ▲ Higher = Better

Smoothed variation of ROE calculating net income relative to the average equity between start and end of the period.

ROAE provides a more accurate metric than simple ROE when a bank undergoes capital increases or major dividend payouts during the year.

ABIX FORMULA:

$$\text{ROAE} = \text{Résultat Net} / ((\text{Fonds Propres } t + \text{Fonds Propres } t-1) / 2) \times 100$$

Interpretation: Delivers an unbiased measure of equity efficiency throughout the entire financial period.

Concrete example: With 15 billion DZD net income and equity rising from 100 to 110 billion: $\text{ROAE} = 15 / 105 = 14.29\%$.

ABIX Modules: Profil Banque, Scores financiers, Comparateur

Return on Assets (ROA) (ROA / RA)

Unit: % | ▲ Higher = Better

Measures how effectively a bank converts its total asset base into net profit.

ROA measures macroeconomic asset efficiency independently of leverage or capitalization structure. In banking, a ROA of 1 to 2% represents high asset productivity.

ABIX FORMULA:

$$\text{ROA} = (\text{Résultat Net} / \text{Total Actifs}) \times 100$$

Interpretation: A high ROA indicates the bank optimizes earnings per balance-sheet unit without relying on unsafe leverage.

Concrete example: Private banks often post higher ROA (1.8% to 2.3%) than large public lenders (1.2% to 1.6%) driven by wider loan spreads.

ABIX Modules: Vue Secteur, Classements, Profil Banque, Heatmap, Scores financiers, Quatre Dimensions

Return on Average Assets (ROAA) (ROAA)

Unit: % | ▲ Higher = Better

Smoothed version of ROA calculating net income over average total assets across the financial year.

ROAA neutralizes intra-year balance sheet expansion to accurately capture average asset productivity.

ABIX FORMULA:

$$\text{ROAA} = \text{Résultat Net} / ((\text{Total Actifs } t + \text{Total Actifs } t-1) / 2) \times 100$$

Interpretation: Preferred indicator by international credit analysts to benchmark banks undergoing rapid asset expansion.

Concrete example: Algerian banking median ROAA stands at approximately 1.52%.

ABIX Modules: Profil Banque, Scores financiers, Compareur

Cost-to-Income Ratio (CIR) (CE / CIR)

Unit: % | ▼ Lower = Better

Percentage of Net Banking Income consumed by operating expenses and depreciation.

The Cost-to-Income ratio is the central benchmark of bank operational efficiency. A lower percentage reflects leaner operations and superior cost discipline.

ABIX FORMULA:

$$\text{Coefficient d'Exploitation} = (\text{Frais Généraux d'Exploitation} + \text{Amortissements}) / \text{Produit Net Bancaire} \times 100$$

Interpretation: Under 45% signifies superior efficiency; 45% to 60% indicates standard cost control; above 65% signals heavy overhead eating into margins.

Concrete example: BEA records a Cost-to-Income ratio of 34.2% (Top Q1), while the market average is 42.6%.

ABIX Modules: Vue Secteur, Classements, Profil Banque, Heatmap, Scores financiers, Quatre Dimensions, Executive Dashboard

Net Interest Margin (NIM) (MIN / NIM)

Unit: % | ▲ Higher = Better

Ratio comparing net interest earnings on loans and bonds against total interest-bearing assets.

NIM measures the core unit profitability of traditional lending and deposit intermediation activities.

ABIX FORMULA:

$$\text{Marge Nette d'Intérêt} = (\text{Produits d'intérêts} - \text{Charges d'intérêts}) / \text{Actifs Productifs Moyens} \times 100$$

Interpretation: A high NIM indicates pricing power on loans or a strong base of zero-cost demand deposits.

Concrete example: Average sector NIM stands around 3.40% in the Algerian market.

ABIX Modules: Vue Secteur, Profil Banque, Heatmap, Scores financiers

NBI per Employee (Productivity) (PNB / Agent)

Unit: M DZD / agent | ▲ Higher = Better

Average Net Banking Income generated per full-time bank employee over the year.

This indicator measures the return on human capital and the degree of branch/back-office automation.

ABIX FORMULA:

$\text{Productivité} = \text{Produit Net Bancaire} / \text{Effectif Total de la Banque (ETP)}$

Interpretation: Private and corporate banks frequently exceed 20-30M DZD/employee due to streamlined headcount and corporate focus.

Concrete example: BEA posts 18.2M DZD NBI/employee (92nd sector percentile) against a market average of 14.8M DZD.

ABIX Modules: Profil Banque, Quatre Dimensions, Scores financiers

RISK, SOLVENCY & PRUDENTIAL

Capital Adequacy Ratio (CAR) (CAR / Cooke)

Unit: % | ▲ Higher = Better

Ratio comparing regulatory capital against the bank's total risk-weighted assets (RWA).

Core international Basel prudential standard implemented by the Bank of Algeria, requiring minimum total regulatory capital above 9.5% - 12% of risk-weighted assets.

ABIX FORMULA:

$\text{Ratio de Solvabilité} = (\text{Fonds Propres Réglementaires} / \text{Actifs Pondérés des Risques - RWA}) \times 100$

Interpretation: A ratio above 15-18% reflects a solid safety buffer supporting credit expansion without capital constraints.

Concrete example: The average capital adequacy ratio across Algerian banks stands at 17.8%, well above regulatory floors.

ABIX Modules: Vue Secteur, Profil Banque, Quatre Dimensions, Scores financiers

Cost of Risk (CoR) (CdR / CoR)

Unit: % | ▼ Lower = Better

Net impairment charges set aside by the bank to cover potential loan defaults relative to total average loan portfolio.

The cost of risk gauges the underlying quality of the credit book and provisioning prudence against non-performing loans.

ABIX FORMULA:

$\text{Coût du Risque} = (\text{Dotations nettes aux provisions pour créances} / \text{Encours moyen de crédits bruts}) \times 100$

Interpretation: A low cost of risk (< 0.5%) reflects clean credit quality. A spike signals rising borrower distress.

Concrete example: BEA posts a controlled cost of risk of 0.65%, whereas specialized agricultural lenders reach 1.45%.

ABIX Modules: Profil Banque, Heatmap, Outliers, Scores financiers, Quatre Dimensions, Executive Dashboard

Non-Performing Loans Ratio (NPL Ratio) (NPL / CDL)

Unit: % | ▼ Lower = Better

Share of loans with delayed repayments or high default probability relative to total gross loans.

The NPL ratio is the standard measure of bank asset quality, encompassing substandard, doubtful, and loss classifications under Bank of Algeria rules.

ABIX FORMULA:

Taux NPL = (Créances Non Performantes / Total des Crédits Bruts) × 100

***Interpretation:** A lower NPL ratio indicates disciplined underwriting and active debt recovery.*

Concrete example: The sector NPL ratio is closely tracked to assess macroeconomic cycle impacts on corporate borrowers.

ABIX Modules: Profil Banque, Quatre Dimensions, Scores financiers

Provision Coverage Ratio (TCP / PCR)

Unit: % | ▲ Higher = Better

Percentage of non-performing loans already covered by accumulated loss provisions on the balance sheet.

This ratio measures balance sheet protection against realized loan defaults: fully provisioned bad debts prevent surprise income hits upon write-off.

ABIX FORMULA:

Taux de Couverture = (Provisions Cumulées pour Dépréciation de Crédits / Total Créances Non Performantes) × 100

***Interpretation:** A coverage ratio above 75-80% reflects prudent balance sheet management and strong loss absorption capacity.*

Concrete example: The average provision coverage ratio across the Algerian banking market stands at 84.2%.

ABIX Modules: Profil Banque, Quatre Dimensions, Scores financiers

Equity-to-Assets Ratio (Capitalization) (FP / Actifs)

Unit: % | ▲ Higher = Better

Proportion of total assets financed directly by equity without borrowing or customer deposits.

Unweighted leverage metric. Unlike risk-weighted CAR, it provides a straightforward, unmodeled view of true capital strength.

ABIX FORMULA:

Ratio FP / Bilan = (Fonds Propres Comptables / Total Bilan) × 100

***Interpretation:** A higher ratio shields depositors and significantly curtails insolvency risk.*

Concrete example: Median Equity-to-Assets ratio across the Algerian banking sector: 8.32%.

ABIX Modules: Vue Secteur, Profil Banque, Quatre Dimensions, Comparateur

Financial Leverage Ratio (Levier / FLR)

Unit: x

Number of times a bank's total assets exceed its equity capital base.

Financial leverage multiplies asset return (ROA) into equity return (ROE). High leverage amplifies gains during good years but heightens insolvency risk under asset deterioration.

ABIX FORMULA:

$$\text{Levier} = \text{Total Bilan} / \text{Fonds Propres} = 1 / (\text{Ratio Fonds Propres} / \text{Bilan})$$

Interpretation: A multiple of 10x to 12x is standard for prudent commercial banking institutions.

Concrete example: For a balance sheet of 1,000B DZD and 100B DZD equity, the leverage multiple is 10.0x.

ABIX Modules: Profil Banque, Quatre Dimensions, Scores financiers

Bank Z-Score (Distance to Default) (Bank Z-Score)

Unit: score | ▲ Higher = Better

Statistical metric measuring how many standard deviations of return decline a bank can withstand before exhausting equity.

The Bank Z-Score combines return on assets (ROA), capitalization (Equity/Assets), and historical return volatility. A higher score reflects greater distance to insolvency.

ABIX FORMULA:

$$\text{Z-Score} = (\text{ROA moyen} + (\text{Fonds Propres} / \text{Total Actifs})) / \text{Écart-type(ROA)}$$

Interpretation: A high Z-score signifies superior earnings stability and stout equity protection against volatility.

Concrete example: Utilized in ABIX advanced benchmarks to evaluate bank stability across macroeconomic cycles.

ABIX Modules: Scores financiers, Profil Banque, Quatre Dimensions

CAMELS Rating Framework (CAMELS)

Unit: cadre

International supervisory framework evaluating banks across 6 pillars: Capital adequacy, Asset quality, Management, Earnings, Liquidity, Sensitivity to market risk.

The CAMELS model serves as the benchmark framework for central bank supervisors and credit rating agencies assessing financial institutions.

ABIX FORMULA:

Évaluation multicritère des 6 dimensions (Capital, Asset quality, Management, Earnings, Liquidity, Sensitivity)

Interpretation: Provides a holistic 360° health check without overemphasizing pure asset volume.

Concrete example: ABIX incorporates the core principles of CAMELS into its 360° Bank Profile and Four Dimensions framework.

ABIX Modules: Profil Banque, Quatre Dimensions, Méthodologie

LIQUIDITY & INTERMEDIATION

Loan-to-Deposit Ratio (LDR) (LDR / RPD)

Unit: %

Percentage of total customer deposits redeployed into customer loans.

LDR measures bank transformation intensity and liquidity cushioning. Below 70-80% reflects excess liquidity, while over 100% indicates the bank relies on external wholesale funding.

ABIX FORMULA:

$$\text{LDR} = (\text{Total Crédits Bruts Clientèle} / \text{Total Dépôts Clientèle}) \times 100$$

Interpretation: In Algeria, the average LDR is moderate (~60.4%), highlighting structural sector liquidity and substantial lending headroom.

Concrete example: Al Baraka posts an LDR of 94.1% (active participatory banking model) compared to 58.2% for BNA.

ABIX Modules: Vue Secteur, Classements, Profil Banque, Heatmap, Outliers, Quatre Dimensions, Executive Dashboard

STATISTICS & QUANTITATIVE METHODS

Compound Annual Growth Rate (CAGR) (CAGR / TCAC)

Unit: % | ▲ Higher = Better

Measures the mean annualized growth rate of a financial metric over a multi-year period, smoothing out year-to-year swings.

CAGR computes the constant annualized rate required for a metric to grow from its initial value to its ending balance over n periods. In ABIX, 3-year and 5-year CAGRs are tracked across balance sheet and P&L; aggregates.

ABIX FORMULA:

$$\text{CAGR} = ((\text{Valeur Finale} / \text{Valeur Initiale})^{(1 / n)}) - 1$$

Interpretation: A CAGR of 8% means the aggregate expanded at an annualized average rate of 8% throughout the analyzed timeframe.

Concrete example: Total Assets expanding from 100B DZD in 2020 to 125.97B DZD in 2023 (3 years): $\text{CAGR} = (125.97 / 100)^{(1/3)} - 1 = +8.0\%$ per annum.

ABIX Modules: Vue Secteur, Historique, Profil Banque, Quatre Dimensions, Executive Dashboard

Year-over-Year Growth (YoY) (YoY)

Unit: % | ▲ Higher = Better

Percentage change in a financial metric compared directly against the previous year's figure.

Measures short-term annual trajectory across financial aggregates from one financial close to the next.

ABIX FORMULA:

$$\text{YoY} = ((\text{Valeur } t - \text{Valeur } t-1) / \text{Valeur } t-1) \times 100$$

Interpretation: Helps identify rapid market accelerations or operational slowdowns between consecutive financial periods.

Concrete example: Total banking assets: +8.4% YoY in 2023.

ABIX Modules: Vue Secteur, Classements, Profil Banque, Historique

Median (50th Percentile / Q2) (Med / Q2)

Unit: selon indicateur

Value dividing an ordered distribution of banks into two equal halves: 50% fall below and 50% above.

In the Algerian banking sector where large state lenders dominate total volume, the median provides a far more representative market benchmark than the arithmetic mean, avoiding distortion from extreme outliers.

ABIX FORMULA:

Valeur au rang central $(N+1)/2$ d'une série ordonnée de N banques

Interpretation: Represents the true central baseline of the domestic banking market.

Concrete example: The median banking ROE stands at 11.8% (differing from the asset-weighted aggregate mean).

ABIX Modules: Vue Secteur, Profil Banque, Matrice Stratégique, Outliers, Quatre Dimensions

Quartiles (Q1, Q2, Q3, Q4) (Q1 / Q2 / Q3 / Q4)

Unit: quartile

Values that split the ordered population of banks into four equal groups, each containing 25% of observations.

Q1 marks the 25th percentile (lower 25% cutoff), Q2 marks the median (50%), and Q3 represents the 75th percentile (top 25% threshold). Q4 denotes top-quartile performers.

ABIX FORMULA:

Q1 = 25e percentile, Q2 = 50e percentile (médiane), Q3 = 75e percentile

Interpretation: Ranking in quartile Q1 for Cost-to-Income means being among the top 25% most efficient institutions (since lower is better).

Concrete example: BEA ranks in the Top 15% (Q4) on ROE and in Q1 (efficient) on Cost-to-Income.

ABIX Modules: Profil Banque, Classements, Heatmap, Scores financiers

Percentile (Centile) (P_x / Centile)

Unit: %ile | ▲ Higher = Better

Indicates the exact percentage of banks in the sector that score below a specific bank's value.

Percentiles position any bank along a normalized 0-to-100 scale across the 21 peer institutions, regardless of the underlying metric scale.

ABIX FORMULA:

Percentile = $(\text{Nombre de banques avec valeur inférieure} / \text{Nombre total de banques}) \times 100$

Interpretation: An 88th percentile ranking signifies the bank outperforms 88% of its peer group on that metric.

Concrete example: On NBI/employee productivity, BEA ranks at the 92nd market percentile.

ABIX Modules: Profil Banque, Classements, Scores financiers

Standard Deviation (Sigma) (σ / Sigma)

Unit: selon indicateur

Measures the dispersion of individual bank values around the sector average.

A larger standard deviation indicates wide performance disparities across institutions. In ABIX, it determines confidence bands and outlier detection thresholds.

ABIX FORMULA:

Écart-type = Racine carrée de la moyenne des écarts au carré par rapport à la moyenne

Interpretation: In a normal distribution, ~68% of banks fall within $\pm 1.0\sigma$ of the mean, and ~95% within $\pm 2.0\sigma$.

Concrete example: Used to calibrate anomaly thresholds on cost of risk or LDR ratios.

ABIX Modules: Outliers, Scores financiers, Méthodologie

Statistical Outlier (Anomaly) (Outlier)

Unit: sigma (σ)

An observation deviating with statistical significance from sector-wide distribution baselines.

ABIX identifies outliers deterministically using Z-Scores ($> \pm 2.0\sigma$ warning, $> \pm 3.0\sigma$ critical) and Interquartile Range (IQR). Outliers often indicate unique business models (e.g., Islamic finance) or one-off shocks.

ABIX FORMULA:

$|Z| = |(X - \text{Moyenne}) / \text{Écart-type}| > 2.0$ ou X hors de $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$

Interpretation: An outlier is not inherently negative: it flags a distinct positioning requiring dedicated qualitative review.

Concrete example: Al Baraka emerges as an LDR outlier ($+3.1\sigma$, 94.1% vs median 64.1%) due to its Islamic financing structure.

ABIX Modules: Outliers, Executive Dashboard, Profil Banque

Standardized Z-Score (Z-Score)

Unit: score

Measures the number of standard deviations a specific value sits above or below the population mean.

Z-Scores allow cross-comparison of financial variables with different units (e.g. DZD, percentages, headcount) onto a unified zero-centered scale.

ABIX FORMULA:

$Z = (\text{Valeur observée} - \text{Moyenne}) / \text{Écart-type}$

Interpretation: A Z of 0 equals the mean; Z of +2.0 places the institution in the upper 2.5% tail.

Concrete example: AGB scores a Z-Score of $+2.6\sigma$ on NBI growth, highlighting notable outperformance.

ABIX Modules: Outliers, Scores financiers, Méthodologie

Min-Max Normalization (0–100 Scale) (Min-Max)

Unit: score 0-100 | ▲ Higher = Better

Mathematical transformation mapping any financial indicator onto a uniform 0 (lowest) to 100 (best) scale.

Used in 360° radar charts and multidimensional matrices to combine disparate criteria into balanced comparable scores.

ABIX FORMULA:

$$\text{Score} = ((\text{Valeur} - \text{Valeur Min}) / (\text{Valeur Max} - \text{Valeur Min})) \times 100$$

Interpretation: Enables immediate intuitive visual assessment of strengths and improvement areas on radar plots.

Concrete example: A score of 100 is assigned to the top-performing bank, and 0 to the lowest.

ABIX Modules: Profil Banque, Quatre Dimensions, Scores financiers

Base 100 Index (Rebasing) (Base 100)

Unit: indice | ▲ Higher = Better

Indexing technique setting a baseline year to 100 to compare relative growth trajectories across banks of differing sizes.

Base 100 indexing plots relative growth speeds of massive state lenders and agile private banks on the same normalized chart.

ABIX FORMULA:

$$\text{Indice Année } t = (\text{Valeur Année } t / \text{Valeur Année de Base}) \times 100$$

Interpretation: An index of 135 in 2024 (base 100 in 2020) represents +35% cumulative growth over the period.

Concrete example: Applied in ABIX Historical and Comparator modules to analyze multi-year growth trajectories.

ABIX Modules: Historique, Comparateur, Vue Secteur

Pearson Correlation Coefficient (r de Pearson)

Unit: [-1 ; +1]

Measures the strength and direction of the linear relationship between two financial variables.

The coefficient ranges from -1 (perfect inverse correlation) to +1 (perfect positive correlation), with 0 indicating no linear relationship.

ABIX FORMULA:

$$r = \text{Covariance}(X, Y) / (\text{Écart-type}(X) \times \text{Écart-type}(Y))$$

Interpretation: Helps empirically verify whether larger bank balance sheets translate into superior profitability margins.

Concrete example: Correlation between Total Assets and ROE is moderate, confirming asset scale alone does not guarantee superior returns.

ABIX Modules: Analyse Transversale, Matrice Stratégique, Méthodologie

SECTOR & MARKET ANALYSIS

Market Share (PDM / MS)

Unit: % | ▲ Higher = Better

Percentage of the total banking industry aggregate represented by an individual bank (Assets, Deposits, Loans, or NBI).

Market share reflects a bank's scale and competitive footprint. ABIX computes market shares across Assets, Deposits, Loans, and NBI, tracking yearly gains and losses.

ABIX FORMULA:
$$\text{Part de Marché} = (\text{Agréгат de la Banque} / \text{Total Agrégat des 21 Banques}) \times 100$$

Interpretation: The 6 state-owned banks hold approx. 88% aggregate asset market share, versus 12% across all private institutions.

Concrete example: BNA (26.8%) and BEA (21.9%) together represent nearly half of the entire Algerian banking market.

ABIX Modules: Parts de marché, Vue Secteur, Profil Banque, Classements, Concentration, Executive Dashboard

Market Share Gain / Loss (Delta) (Δ PDM)

Unit: pts de % | ▲ Higher = Better

Difference in percentage points between a bank's market share in the current year versus the previous year.

Identifies which institutions gain or lose market traction in deposits, lending, or assets, independent of overall market expansion.

ABIX FORMULA:
$$\text{Variation PDM} = \text{Part de Marché } t - \text{Part de Marché } t-1 \text{ (en points de \%)}$$

Interpretation: A gain of +0.82 percentage points denotes growth outpacing the broader market average.

Concrete example: BEA gained +0.82 points of total asset market share in 2022, the strongest gain in the sector.

ABIX Modules: Parts de marché, Vue Secteur, Profil Banque, Executive Dashboard

Peer Group (Peer Group)

Unit: segment

Subset of banks sharing homogeneous characteristics (size, public/private ownership, business model) for meaningful benchmarking.

Comparing a mid-sized private lender directly against public mega-banks can skew insights. ABIX isolates relevant peer cohorts (Public banks, International private banks, Islamic banks) for objective relative evaluation.

ABIX FORMULA:
Segmentation par critères : Statut (Public/Privé), Taille (Top 5 / Intermédiaires), Modèle (Universel / Islamique)

Interpretation: Evaluates whether a bank outperforms direct competitors operating under identical regulatory and market conditions.

Concrete example: Al Baraka is benchmarked against its specific peer cohort of Islamic and private banks.

ABIX Modules: Comparateur, Profil Banque, Quatre Dimensions, Executive Dashboard

Banking Cluster (Strategic Group) (Cluster)

Unit: typologie

Algorithmic grouping of banks displaying similar multidimensional financial profiles (profitability, leverage, efficiency, growth).

ABIX identifies distinct structural clusters in Algeria: 1) Large universal public institutions, 2) High-profitability private lenders, 3) Growth-oriented expansion banks, 4) Niche/Islamic institutions.

ABIX FORMULA:
Classification par analyse en composantes principales (ACP) et K-Means sur les ratios clés

Interpretation: Highlights the rich diversity of operating models across the Algerian banking landscape.

Concrete example: Visualized on the ABIX Growth vs Profitability Strategic Matrix.

ABIX Modules: Matrice Stratégique, Profil Banque, Analyse Transversale

Observed Scope (Sector Coverage) (Scope)

Unit: banques

Total set of licensed commercial banks with published certified financial statements included in the study.

ABIX covers the entirety of commercial banks operating in Algeria: 20 banks in the 2025 edition (2023-2024 data) and 21 banks in the 2026 edition (including Ziraat Bankasi).

ABIX FORMULA:
Périmètre = 6 Banques Publiques + 15 Banques Privées / Mixtes = 21 Banques

Interpretation: Ensures comprehensive representative coverage of the Algerian market without sampling bias.

Concrete example: 21 banks analyzed covering 100% of commercial banking assets in Algeria.

ABIX Modules: Vue Secteur, Méthodologie, Data Explorer, Classements

Constant Scope (Like-for-Like Scope) (LFL)

Unit: méthode

Multi-year analysis method retaining the identical bank cohort across years to avoid distortion from market entries or exits.

Guarantees measured sector growth is organic rather than an artifact of adding a new bank into the dataset.

ABIX FORMULA:
Agrégat Constant = Somme des grandeurs des banques présentes à la fois en Année t et en Année t-1

Interpretation: Ensures scientific rigor in ABIX multi-year trend comparisons.

Concrete example: Applied when computing sector-wide growth rates over 2022–2025.

ABIX Modules: Historique, Méthodologie, Vue Secteur

Public vs Private Banking Segmentation (PUB / PRIV)

Unit: % de part

Structural comparison of performances, balance sheets, and operating models across state-owned versus private banks.

Algeria features a dual banking landscape: 6 state lenders hold ~88% of assets through dense branch networks, while 15 private banks deliver higher unit profitability (ROE, ROA) on selective customer segments.

ABIX FORMULA:

Agrégat Public = $\sum$ Banques Publiques | Agrégat Privé = $\sum$ Banques Privées

Interpretation: Highlights complementarity between public balance sheet scale and private operational agility.

Concrete example: Private banks maintain an equity return edge (ROE 14.6% vs 11.4% public) despite running fewer branches.

ABIX Modules: Vue Secteur, Parts de marché, Profil Banque, Executive Dashboard

CONCENTRATION & COMPETITION

Herfindahl-Hirschman Index (HHI) (HHI)

Unit: points / 10 000

Global benchmark metric measuring market concentration by summing the squared market shares of all competing banks.

HHI ranges from 0 to 10,000 points. International standards classify: 1) < 1,500 = Unconcentrated, 2) 1,500 - 2,500 = Moderately Concentrated, 3) > 2,500 = Highly Concentrated.

ABIX FORMULA:

HHI = Somme des (Part de marché de chaque banque en %)²

Interpretation: With an asset HHI around 1,742 points, the Algerian banking sector falls into the moderately concentrated tier, shaped around a sturdy public core.

Concrete example: BNA (26.8% -> 718 pts) + BEA (21.9% -> 480 pts) + CPA (15.2% -> 231 pts) contribute over 80% of the aggregate HHI.

ABIX Modules: Concentration, Vue Secteur, Data Explorer, Executive Dashboard

Effective Number of Banks (Numbers Equivalent) (N_eff)

Unit: banques | ▲ Higher = Better

Theoretical number of equal-sized banks that would generate the exact same HHI concentration as the current market.

Translates abstract HHI points into an intuitive count: while 21 licensed banks operate, market share skew means the market economically behaves like ~5.7 equal-sized banks.

ABIX FORMULA:

Nombre Effectif = 10 000 / HHI

Interpretation: The closer to 21, the more balanced the market. A figure of 5.7 captures the scale asymmetry between mega-lenders and boutique private banks.

Concrete example: For an HHI of 1,742, the effective number of banks equals 10,000 / 1,742 = 5.74 equivalent banks.

ABIX Modules: Concentration, Vue Secteur, Executive Dashboard

Top 3 Concentration Ratio (CR3) (CR3)

Unit: %

Cumulative market share held by the top 3 banking institutions (BNA + BEA + CPA).

Measures top-tier oligopoly concentration. A CR3 above 60% indicates a tight oligopoly where the top 3 leaders anchor systemic liquidity and pricing.

ABIX FORMULA:

CR3 = Part de Marché Rang 1 + Part de Marché Rang 2 + Part de Marché Rang 3

Interpretation: In Algeria, CR3 reaches 63.9% of total assets, highlighting the dominant footprint of the top 3 state banks.

Concrete example: BNA (26.8%) + BEA (21.9%) + CPA (15.2%) = 63.9% of total assets.

ABIX Modules: Concentration, Vue Secteur, Executive Dashboard

Top 5 Concentration Ratio (CR5) (CR5)

Unit: %

Cumulative market share held by the top 5 banking institutions (BNA, BEA, CPA, BADR, CNEP).

CR5 gauges the combined market dominance of the primary Tier-1 banking group in financing the national economy.

ABIX FORMULA:

CR5 = Somme des parts de marché des 5 premières banques

Interpretation: CR5 reaches 82.0% of total assets in Algeria, showing that Tier-1 state banks concentrate the vast bulk of banking resources.

Concrete example: The top 5 banks assemble over 18,000B DZD out of the 22,000B DZD total industry assets.

ABIX Modules: Concentration, Vue Secteur, Executive Dashboard

ABIX METHODOLOGY & FRAMEWORK

ABIX Four Dimensions Framework (4D ABIX)

Unit: cadre

Transparent benchmarking framework avoiding arbitrary global scores, structured around 4 distinct pillars: 1. Profitability, 2. Growth, 3. Efficiency, 4. Capitalization.

ABIX avoids opaque single-score rankings. The 4D framework delivers an objective multidimensional read where institutions are benchmarked on verifiable metrics.

ABIX FORMULA:

Piliers : 1. Rentabilité (ROE, ROA, NIM) | 2. Croissance (CAGR Actifs, Dépôts, PNB) | 3. Efficacité (CIR, PNB/Agent) | 4. Capitalisation (FP/Actifs, Solvabilité)

Interpretation: Allows executive leadership and analysts to quickly grasp an institution's distinct profile without methodology shortcuts.

Concrete example: A bank may excel in profitability while remaining in consolidation mode regarding growth.

ABIX Modules: Quatre Dimensions, Profil Banque, Méthodologie, Data Explorer

Growth vs Profitability Matrix (4 Quadrants) (Matrice 2D)

Unit: matrice

Visual positioning cross-referencing NBI growth rate (X-axis) and ROE equity profitability (Y-axis) against sector medians.

The matrix segments banks into 4 quadrants: 1) Dynamic Leaders (High growth, High ROE), 2) Resilient Profitability (Moderate growth, High ROE), 3) Investment & Expansion (High growth, Moderate ROE), 4) In Transformation (Moderate growth, Moderate ROE).

ABIX FORMULA:

Axes centrés sur la Médiane Secteur : Axe X = Croissance PNB % | Axe Y = Rentabilité ROE % | Taille bulle = Total Bilan

Interpretation: Provides an instant snapshot of commercial momentum and financial return across every banking player.

Concrete example: BEA and Al Baraka position inside the Dynamic Leaders quadrant.

ABIX Modules: Matrice Stratégique, Profil Banque, Executive Dashboard

Financial Transparency & Availability Score (Transparence)

Unit: / 6 pts | ▲ Higher = Better

Objective score evaluating the accessibility and timeliness of audited financial statements and annual reports on official bank websites.

Structured on 2 dimensions (Financial Statements publication /3 pts and Annual Activity Report /3 pts) for a total global score of 6 points.

ABIX FORMULA:

Score Global (0 à 6) = Score Publication des Comptes (0 à 3) + Score Rapport d'Activité (0 à 3)

Interpretation: Measures adherence to best practices in public financial disclosure and investor relations.

Concrete example: 3 banks achieve the perfect score of 6/6 (CPA, BDL, ABC), while the market average sits at 2.7/6.

ABIX Modules: Disponibilité Financière, Méthodologie

Deterministic Scanning & Strategic Alerts (Scanning ABIX)

Unit: alertes

Algorithmic engine automatically scanning all 21 balance sheets to detect trend breaks, structural divergences, and executive signals.

Unlike black-box AI generative models, ABIX strategic alerts are 100% deterministic, auditable, and directly traceable to certified published financial line items.

ABIX FORMULA:

Règles déterministes : Divergence Dépôts/Crédits, Rupture de Marge, Écart de Provisionnement ($> \pm 2\sigma$), Érosion PNB vs Résultat

Interpretation: Supplies executive committees and boards with an instant, verified briefing on key structural market developments.

Concrete example: Detected alert: sector deposits (+6.2%) expanded faster than loans (+5.1%), widening the aggregate liquidity cushion.

ABIX Modules: Executive Dashboard, Vue Secteur, Profil Banque

Digital Banking & E-Banking Barometer (Baromètre)

Unit: baromètre

Periodic market survey assessing customer adoption, digital usage, and user satisfaction across web, mobile banking, and branch networks.

Operational complement to balance sheet data, the Barometer links certified financial performance with real-world digital customer adoption in Algeria.

ABIX FORMULA:

Enquête quantitative et qualitative : Canaux digitaux, Ergonomie, Disponibilité, Tarification, Sécurité

Interpretation: Sheds light on customer digital expectations and technological readiness across Algerian institutions.

Concrete example: 2026 Edition: in-depth comparative benchmark of mobile banking app features in Algeria.

ABIX Modules: Baromètre 2026, Méthodologie, Édition 2026